1. DEFINITIONS AND NOMENCLATURE

Defining and describing the severity of PAD

The term “critical limb ischemia” (CLI) is outdated and fails to encompass the full spectrum of patients who are evaluated and treated for limb-threatening ischemia in modern practice. Instead, the new term CLTI is proposed to include a broader and more heterogeneous group of patients with varying degrees of ischemia that can often delay wound healing and increase amputation risk.

For development of a clearer concept of CLTI, the following are excluded from the population as defined in this guidelines document: patients with purely venous ulcers, acute limb ischemia, acute trash foot, ischemia due to emboli, acute trauma, or mangled extremity and those with wounds related to nonatherosclerotic conditions. These include vasculitides, collagen vascular disease, Buerger’s disease, neoplastic disease, dermatoses, and radiation arteritis.

Previous leg ischemia definition and classification systems

CLI. In 1982, a working group of vascular surgeons defined CLI as ischemic rest pain with an ankle pressure (AP) <40 mm Hg, or tissue necrosis with an AP <60 mm Hg, in patients without diabetes.[144] Patients with diabetes were specifically excluded because of the confounding effects of neuropathy and susceptibility to infection. This definition has long been debated because it failed to capture a large group of patients who were at risk for amputation from a broader range of ischemia.[145,146] To address this limitation, multiple and disparate lower limb ischemia and wound/DFU classification systems have been developed and promulgated during the past 5 decades, many of which remain in use today. These and other commonly used classifications and their associated components and grades of severity are summarized in Table 1.1.[10,147-158] Among vascular surgeons, the Fontaine and Rutherford classifications have been the most widely adopted, whereas orthopedists, podiatric surgeons, and diabetic foot specialists traditionally applied the Wagner and University of Texas classifications. The strengths and limitations of each have been widely discussed in previous key publications.[10,150,159-161] Although each of these systems has advantages, the use of multiple classification systems has hindered the development of optimal treatment algorithms. It has also contributed to the fragmentation and variability of care provided for patients with DFUs as well as for nondiabetic patients across the spectrum of CLTI.

Lower extremity threatened limb classification system

The definitions summarized in Table 1.1 were developed primarily to describe patients suffering from pure ischemia due to atherosclerosis. This was when the predominant risk factor was tobacco smoking and before the global epidemic of diabetes mellitus (DM). As such, these definitions were ischemia-dominant models of limb threat. However, because patients with DM now make up the majority of patients with CLTI, absolute perfusion now needs to be considered in the context of neuropathy, wound characteristics, and infection. To address this unmet need, the SVS Lower Extremity Guidelines Committee created the SVS Lower Extremity Threatened Limb Classification System. This system stratifies amputation risk according to wound extent, degree of ischemia, and presence and severity of foot infection (Wound, Ischemia, and foot Infection [WIfI]).[10] Although it may require some adjustments, WIfI appears to correlate strongly with important clinical outcomes. This includes those set forth in the SVS objective performance goals (OPGs) that focus on limb amputation, 1-year amputation-free survival (AFS), and wound healing time (Table 1.2).[10,68-72,162-167]

The WIfI classification system is currently being evaluated in multicenter trials including the U.S. National Institutes of Health-funded BEST-CLI trial[13] and the UK National Institute for Health Research Health Technology Assessment-funded BASIL-2 and BASIL-3 trials.[14,15] WIfI is also being incorporated into the U.S. SVS Vascular Quality Initiative registry of lower extremity interventions.

Hemodynamic criteria

Although previous guidelines have suggested a range of AP and toe pressure (TP) thresholds for defining limb-threatening ischemia, such thresholds must be used with great caution and considered in the clinical context because of multiple confounding factors and the lack of a clear and reliable relationship to outcomes. Patients with limb-threatening ischemia should be defined primarily in terms of their clinical presentation, supplemented by physiologic studies that demonstrate a degree of ischemia sufficient to cause pain, to impair wound healing, and to increase amputation risk.

In addition to patients who meet the proposed new definition of CLTI, there are a significant number of patients whose PAD is so severe that they are likely to be at increased risk for development of CLTI in the foreseeable future.[168] Although data are lacking, it is logical to suggest that such individuals should be monitored closely for clinical disease progression.

CLTI

We propose that CLTI be defined to include a broader and more heterogeneous group of patients with varying degrees of ischemia that may delay wound healing and increase amputation risk. A diagnosis of CLTI requires objectively documented atherosclerotic PAD in association with ischemic rest pain or tissue loss (ulceration or gangrene).

Ischemic rest pain is typically described as affecting the forefoot and is often made worse with recumbency while being relieved by dependency. It should be present for >2 weeks and be associated with one or more abnormal hemodynamic parameters. These parameters include an ankle-brachial index (ABI) <0.4 (using higher of the dorsalis pedis [DP] and posterior tibial [PT] arteries), absolute highest AP <50 mm Hg, absolute TP <30 mm Hg, transcutaneous

Table 1.1. Classification schemes used for chronic limb ischemia and ulceration

Classification system	Ischemic rest pain	Ulcer	Gangrene	Ischemia	Infection	Key features and comments
Ischemia and PAD classifications						
Fontaine (1954)	Yes (class III/IV)	Class IV/IV; ulcer and gangrene grouped together	Class IV/IV; ulcer and gangrene grouped together	Cutoff values for CLI based on European consensus document: Ischemic rest pain >2 weeks with AP <50 mm Hg or TP <30 mm Hg Ulcer and gangrene: AP <50 mm Hg, TP <30 mm Hg, absent pedal pulses in patient with diabetes	No	Pure ischemia model No clear definitions of spectrum of hemodynamics; minimal description of wounds; infection omitted
Rutherford (1997)	Yes (category 4/6)	Category 5: minor tissue loss, nonhealing ulcer, focal gangrene with diffuse pedal ischemia	Category 6: major tissue loss extending above TM level, functional foot no longer salvageable (although, in practice, often refers to extensive gangrene, potentially salvageable foot with significant efforts)	Yes; cutoffs for CLI Category 4: resting AP <40 mm Hg; flat or barely pulsatile ankle or forefoot PVR; TP <30 mm Hg Category 5/6: AP <60 mm Hg; flat or barely pulsatile ankle or forefoot PVR; TP <40 mm Hg	No	Pure ischemia model PAD classification system includes milder forms of PAD (categories 1-3). Categories 4-6 based on cutoff values for CLI; no spectrum of ischemia, does not acknowledge potential need for revascularization, with CLI cutoff depending on wound extent/infection; not intended for patients with diabetes; wound classes not sufficiently detailed; omits infection as a trigger
Second European Consensus (1991)	Yes; pain >2 weeks requiring analgesia; AP ≤50 mm Hg or TP ≤30 mm Hg	Yes, if AP ≤50 mm Hg or TP ≤30 mm Hg	Yes, if AP ≤50 mm Hg or TP ≤30 mm Hg	One hemodynamic cutoff for ulcer and gangrene, with or without diabetes	No	Ischemia threshold too low, especially for patients with diabetes; wounds not graded; infection not considered
TASC I (2000)	Yes, if ischemia criteria met	Yes, if ischemia criteria met	Yes, if ischemia criteria met	One hemodynamic cutoff, with no differentiation of diabetics from nondiabetics	No	Focused primarily on arteriographic anatomy without detailed stratification of the limb itself (wounds and infection not graded)
TASC II (2007)	Yes, if AP <50 mm Hg or TP <30 mm Hg	Yes, if ischemia criteria met of AP <70 mm Hg or TP <50 mm Hg	Yes, if ischemia criteria met of AP <70 mm Hg or TP <50 mm Hg	Yes, but noted "there is not complete consensus regarding the vascular haemodynamic parameters required to make the diagnosis of CLI"	No	Focused primarily on arteriographic anatomy without detailed stratification of the limb itself (wounds and infection not graded); issues with hemodynamic criteria noted
DFU classifications						
Meggitt-Wagner (1976, 1981)	No	Grade 0: pre- or post-ulcerative lesion Grade 1: partial/full-thickness ulcer Grade 2: probing to tendon or capsule Grade 3: deep ulcer with osteitis Grade 4: partial foot gangrene Grade 5: whole foot gangrene	Ulcer and gangrene grouped together; gangrene due to infection not differentiated from gangrene due to ischemia; also includes osteomyelitis	No	No for soft tissue component; included only as osteomyelitis	Orthopedic classification intended for diabetic feet No hemodynamics; gangrene from infection not differentiated from that due to ischemia; osteomyelitis included; soft tissue infection not separated from bone infection

Table 1.1-*continued*

Classification system	Ischemic rest pain	Ulcer	Gangrene	Ischemia	Infection	Key features and comments
University of Texas (1998)	No	Yes: grade 0-III ulcers Grade 0: pre- or post-ulcerative completely epithelialized lesion Grade I: superficial, not involving tendon, capsule, or bone Grade II: penetrating to tendon/capsule Grade III: penetrating to bone or joint	No	Yes: binary ± based on ABI <0.8	Yes ± wounds, with frank purulence or >2 of the following (warmth, erythema, lymphangitis, edema, lymphadenopathy, pain, loss of function) considered infected	Primarily intended for DFUs; includes validated ulcer categories; PAD and infection included, but only as ± variable with no grades/spectrum
S(AD) SAD system (1999)	No	Yes: 0-3 based on area and depth Grade 0: skin intact Grade 1: superficial, $<$ 1 cm2 Grade 2: penetrates to tendon, periosteum, joint capsule, 1-3 cm2 Grade 3: lesions in bone or joint space, >3 cm2	No	Pulse palpation only, no objective hemodynamic testing	Yes; 1 = no infection, 2 = cellulitis, 3 = osteomyelitis	Intended for DFUs; also includes neuropathy; does not mention gangrene; no hemodynamic information, perfusion assessment based on pulse palpation only
PEDIS (2004)	No	Yes: grades 1-3 Grade 1: superficial full-thickness ulcer, not penetrating deeper than the dermis Grade 2: deep ulcer, penetrating below the dermis to subcutaneous structures involving fascia, muscle, or tendon Grade 3: all subsequent layers of the foot involved including bone and joint (exposed bone, probing to bone)	No	Yes: 3 grades, CLI cutoff Grade 1: no PAD symptoms, ABI >0.9, TBI >0.6, $TcPO_2$ >60 mm Hg Grade 2: PAD symptoms, ABI <0.9, AP >50 mm Hg, TP >30 mm Hg, $TcPO_2$ 30-60 mm Hg Grade 3: AP <50 mm Hg, TP <30 mm Hg, $TcPO_2$ <30 mm Hg	Yes: grades 1-4 based on IDSA classification	Primarily intended for DFUs; ulcer grades validated; includes perfusion assessment, but with cutoff for CLI; gangrene not separately categorized; includes validated IDSA infection categories
Saint Elian (2010)	No	Yes: grades 1-3 based on depth Grade 1: superficial wound disrupting entire skin Grade 2: moderate or partial depth, down to fascia, tendon, or muscle but not bone or joints Grade 3: severe or total, wounds with bone or joint involvement Multiple categories including area, ulcer number, location, and topography	No	Yes: grades 0-3 Grade 0: AP >80 mm Hg, ABI 0.9-1.2 Grade 1: AP 70-80 mm Hg, ABI 0.7-0.89, TP 55-80 mm Hg Grade 2: AP 55-69 mm Hg, ABI 0.5-0.69, TP 30-54 mm Hg Grade 3: AP <55 mm Hg, ABI <0.5, TP <30 mm Hg	Yes: grades 0-3 Grade 0: none Grade 1: mild; erythema 0.5-2 cm, induration, tenderness, warmth, and purulence Grade 2: moderate; erythema >2 cm, abscess, muscle tendon, joint, or bone infection Grade 3: severe; systemic response (similar to IDSA)	Detailed system intended only for DFUs; comprehensive ulcer classification system with hemodynamic categories for gradations of ischemia; gangrene not considered separately Infection system similar to IDSA
IDSA (2012)	No	No	No	No	Yes: uninfected, mild, moderate, and severe	Validated system for risk of amputation related to foot infection but not designed to address wound depth/ complexity or degree of ischemia

Continued

Table 1.1-*continued*

Classification system	Ischemic rest pain	Ulcer	Gangrene	Ischemia	Infection	Key features and comments
Recommended CLTI classification						
SVS WIfI threatened limb classification (2014)	Yes, if confirmed by hemodynamic criteria	Yes: grades 0-3 Grouped by depth, location, and size and magnitude of ablative/ wound coverage procedure required to achieve healing	Yes: grades 0-3 Grouped by extent, location, and size and magnitude of ablative or wound coverage procedure required to achieve healing	Yes: ischemia grades 0-3 Hemodynamics with spectrum of perfusion abnormalities; no cutoff value for CLI Grade 0 unlikely to require revascularization	Yes: IDSA system (grades 0-3); grades corelate with amputation risk	Includes PAD $\pm$ diabetes with a range of wounds, ischemia, and infection, scaled from 0-3 No single cutoff for CLI as CLTI is considered a spectrum of disease Need for revascularization depends on degree of ischemia, wound, and infection severity Ulcers/gangrene categorized by extent and complexity of anticipated ablative surgery/coverage

ABI = Ankle-brachial index; AP = ankle pressure; CLI = critical limb ischemia; DFU = diabetic foot ulcer; CLTI = chronic limb-threatening ischemia; IDSA = Infectious Diseases Society of America; PAD = peripheral artery disease; PEDIS = perfusion, extent, depth, infection, and sensation; PVR = pulse volume recording; SVS = Society for Vascular Surgery; TASC = TransAtlantic Inter-Society Consensus; TBI = toe-brachial index; $TcPO_2$ = transcutaneous oximetry; TM = transmetatarsal; TP = toe pressure; WIfI = Wound, Ischemia, foot Infection.

partial pressure of oxygen ($TcPO_2$) <30 mm Hg, and flat or minimally pulsatile pulse volume recording (PVR) waveforms (equivalent to WIfI ischemia grade 3). Pressure measurements should be correlated with Doppler arterial waveforms, keeping in mind that AP and ABI are frequently falsely elevated because of medial calcinosis, especially in people with DM and end-stage renal disease (ESRD). For this reason, a combination of tests may be needed. In patients with DM or ESRD, toe waveforms and systolic pressures are preferred. One study demonstrated that AP alone failed to identify 42% of patients with CLTI. TP and $TcPO_2$ measurements were more accurate than AP and also were more predictive of 1-year amputation risk (TP <30 mm Hg or $TcPO_2$ <10 mm Hg).[169]

Tissue loss related to CLTI includes gangrene of any part of the foot or nonhealing ulceration present for at least 2 weeks. It should be accompanied by objective evidence of significant PAD (eg, WIfI ischemia grade $\geq$1). This definition excludes purely neuropathic, traumatic, or venous ulcers lacking any ischemic component. However, the WIfI scheme recognizes that a wide range of ischemic deficit may be limb threatening when it coexists with varying degrees of wound complexity and superimposed infection. CLTI is present if either ischemic rest pain or tissue loss with appropriate hemodynamics is present.

Some patients may have relatively normal hemodynamics when the limb or foot is considered as a whole but nevertheless suffer ulceration as a result of diminished local perfusion (ie, angiosomal or regional ischemia without adequate collateral flow). It is recognized that such ulcers may contribute to limb threat, and current tools to assess regional ischemia require further development to better define such circumstances and their treatment. The relationship between regional ischemia and patterns of IP and pedal disease also requires more in-depth study.[12,170]

The GVG recommends use of the SVS WIfI classification (Section 3) in a manner analogous to the TNM system of

Table 1.2. One-year major limb amputation rate by Society for Vascular Surgery (SVS) Wound, Ischemia, and foot Infection (WIfI) clinical stage

Study (year): No. of limbs at risk	Stage 1	Stage 2	Stage 3	Stage 4
Cull[68] (2014): 151	37 (3)	63 (10)	43 (23)	8 (40)
Zhan[69] (2015): 201	39 (0)	50 (0)	53 (8)	59 (64)[a]
Darling[71] (2016): 551	5 (0)	110 (10)	222 (11)	213 (24)
Causey[70] (2016): 160	21 (0)	48 (8)	42 (5)	49 (20)
Beropoulis[163] (2016): 126	29 (13)	42 (19)	29 (19)	26 (38)
Ward[166] (2017): 98	5 (0)	21 (14)	14 (21)	58 (34)
Darling[164] (2017): 992	12 (0)	293 (4)	249 (4)	438 (21)
Robinson[72] (2017): 280	48 (2.1)	67 (7.5)	64 (7.8)	83 (17)
Mathioudakis[165] (2017): 217	95 (4)	33 (3)	87 (5)	64 (6)
Tokuda[167] (2018): 163	16 (0)	30 (10)	56 (10.7)	61 (34.4)
N = 2982 (weighted mean)	**307 (3.2)**	**757 (7.0)**	**859 (8.7)**	**1059 (23.3)**
Median (1-year major limb amputation)	**0%**	**9%**	**9.4%**	**29%**

The number of limbs at risk in each WIfI stage is given, with percentage of amputations at 1 year in parentheses. Means in totals (in parentheses) are weighted.

[a] Falsely elevated because of inadvertent inclusion of stage 5 (unsalvageable) limbs.

cancer staging to stage the limb in patients with CLTI. The WIfI classification is intuitive and has been made user-friendly by the availability of free online application software provided by the SVS (SVS Interactive Practice Guidelines; https://itunes.apple.com/app/id1014644425).

Data accrued in nearly 3000 patients to date and summarized in Table 1.2 suggest that the four WIfI clinical stages of limb threat correlate with the risk of major limb amputation and time to wound healing. It has also been suggested that novel WIfI composite and mean scores may predict other clinically significant events as well.[164] The WIfI system appears to contain the key limb status elements needed to gauge the severity of limb threat at presentation.

In addition, recent data suggest that WIfI can assist in predicting which patients might fare better with open surgical bypass compared with endovascular therapy.[171,172] One study reported that when endovascular therapy alone was applied to WIfI stage 4 patients, results were worse than in lower clinical stage patients.[172] Specifically, the wound healing rate was only 44%, the major limb amputation rate was 20%, and 46% of patients required multiple, repetitive endovascular procedures. In a nonrandomized, single-center comparison of WIfI stage 4 patients, researchers found that freedom from major limb amputation was superior in patients who underwent bypass compared with those who underwent endovascular therapy.[171] If these results can be confirmed, WIfI may prove to be a useful tool in deciding whether to offer endovascular therapy or bypass.

Another study used WIfI in a fashion analogous to TNM staging for cancer and reassigned patients to stages after 1 month of therapy. The investigators found that at 1 month and 6 months, wound, ischemia, and infection grades correlated with AFS, whereas baseline ischemia grade did not.[173] These data suggest that restaging with WIfI at 1 month and 6 months after intervention may help identify a cohort of patients undergoing therapy for CLTI that remains at higher risk for major limb amputation and may merit targeted reintervention.

Ultimately, the optimal staging system for CLTI is expected to evolve with additional clinical application and larger scale, multicenter, and multinational data analysis.

Recommendations 1

1.1 Use objective hemodynamic tests to determine the presence and to quantify the severity of ischemia in all patients with suspected CLTI.

Grade	Level of evidence	Key references
1 (Strong)	C (Low)	de Graaff,[16] 2003 Brownrigg,[17] 2016 Wang,[18] 2016

1.2 Use a lower extremity threatened limb classification staging system (eg, SVS's WIfI classification system) that grades wound extent, degree of ischemia, and severity of infection to guide clinical management in all patients with suspected CLTI.

Grade	Level of evidence	Key references
1 (Strong)	C (Low)	See Table 1.2

2. GLOBAL EPIDEMIOLOGY AND RISK FACTORS FOR CLTI

In 2010, estimates suggested that >200 million people worldwide were living with PAD. This represented a 23.5% increase since 2000, an increase that is believed to be largely attributable to aging populations and the growing prevalence of risk factors, in particular DM.[1] These figures are thought to almost certainly underestimate the true burden of disease as they are largely based on community-based studies that define PAD on the basis of reduced ABI. Although CLTI is widely believed to be a growing global health care problem, reliable epidemiologic data are extremely limited.

Men have been reported to have a higher prevalence of PAD in high-income countries (HICs; Fig 2.1), whereas women seem to have a higher prevalence of PAD in low- and middle-income countries (LMICs).[1] As life expectancy increases, the burden of PAD seems likely to rise in LMIC. However, in certain geographic regions, notably in the western Pacific and Southeast Asia, most PAD cases are reported in people younger than 55 years.[1]

In a meta-analysis from the United States, the prevalence of PAD in men ranged from 6.5% (aged 60-69 years) to 11.6% (aged 70-79 years) to 29.4% (>80 years).[174] There were similar age-related increases in PAD prevalence in women (5.3%, 11.5%, and 24.7% in these age categories, respectively).[174] Given that the life expectancy of women still exceeds that of men, the overall burden of PAD (total number of individuals affected) is likely to be greater in women than in men. The epidemiology of PAD is likely to be similar in other developed countries, such as the United Kingdom, and regions, such as the European Union.[175,176] However, as these populations become more multicultural, differences in disease burden between different communities within these nations seem likely to become apparent, further complicating the epidemiology of the condition.[177]

Data on the epidemiology of PAD and in particular of CLTI in other parts of the world are even more limited. In one Japanese community study of people older than 40 years, the prevalence of ABI <0.9 was very low (1.4%).[178] In a population-based cohort of 4055 Chinese men and women older than 60 years, the prevalence of PAD (ABI <0.9) was 2.9% and 2.8%, respectively.[179] Another population-based cohort of 1871 individuals younger than 65 years in two countries from Central Africa showed that the overall prevalence of PAD was 14.8%.[180]

There is a considerable body of evidence showing that PAD is more common among black individuals than among whites.[181-184] There is also evidence that Asians and Hispanics have a lower prevalence of PAD than whites do.[184] It is not clear whether these differences have a genetic basis or simply reflect differential exposure to traditional risk factors. However, disease risk profiles appear to change as populations migrate, suggesting that environment is more important than genetic makeup. Another explanation may be that ABI is intrinsically lower in black individuals, resulting in a falsely high prevalence of PAD.[185]